

Exhibit Document — “Founders as the Next Offset”

Deliverable 1. Three exhibits supporting the argument that America’s founder class is its decisive asset in the techno-economic competition with China. Each entry contains the exhibit, a plain-English annotation written for a smart non-economist, and a full source citation. Figures are reproducible from the scripts in `code/`; see the repository README.

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Supporting materials. Full code, data, and step-by-step reproduction instructions are in the project’s GitHub repository: github.com/felix-aidala/Cardinal40-Technical. The complete log of substantive AI-tool interactions — how each tool was used and how AI-generated statistics and citations were independently verified — is at `logs/ai_interaction_log.md`.

The three exhibits trace one logical chain:

1. **Where frontier talent now works** — innovation has moved from the academy into industry and company-building.
2. **Why the founder specifically matters** — founders are not interchangeable managers; remove them and innovation measurably falls.
3. **Why America’s system wins** — the U.S. funds its founders at a scale no peer economy approaches, which is why it (not its allies) is the one equipped to challenge China.

Exhibit 1 — Since the 1970s, industry has caught up with academia for new U.S. PhDs

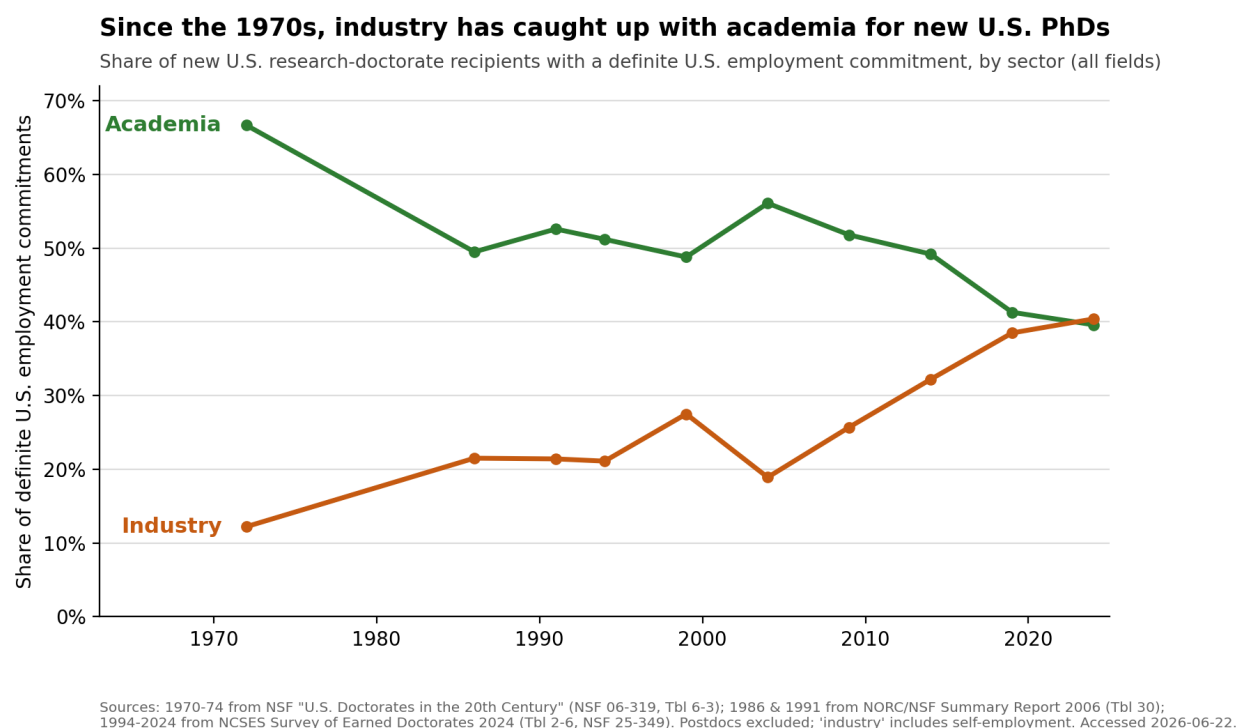


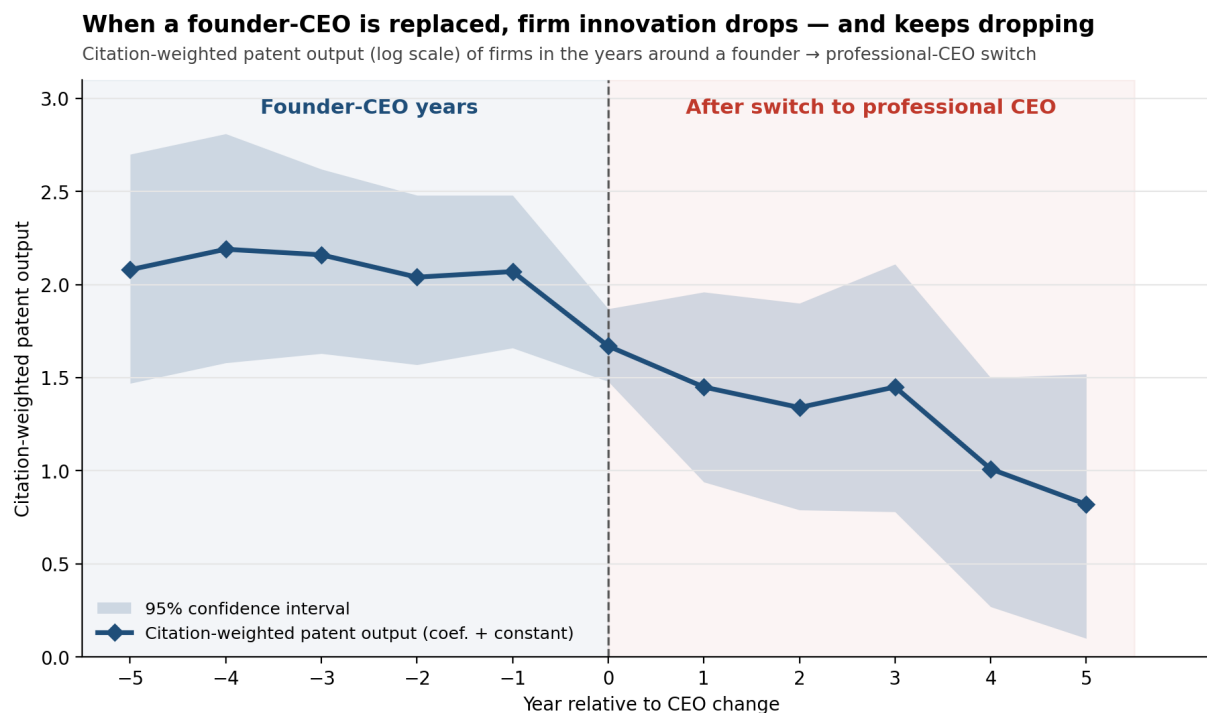
Figure 1: Exhibit 1

Annotation. This figure anchors the essay’s opening claim: that the locus of American technological advantage has shifted since the Second World War and the Cold War that followed. The scientists who

built the first nuclear weapons were largely academics, working for the federal government on leave from their universities; today the equivalent talent — the country’s most capable new PhDs — increasingly leaves the academy for industry, where it builds and leads America’s frontier companies (Anthropic among them). That steady migration of scientific talent into the private sector is what the figure is meant to show, and it is what motivates the investment thesis that follows. Two cautions are worth stating plainly. First, to recover a picture reaching this far back I had to use all fields rather than science and engineering alone, so these are not exclusively technical PhDs. Second, the chart shows talent moving *into industry*, not talent *becoming founders* — I am careful not to imply the latter, because the great majority of these scientists never start a company. The claim is the narrower and more defensible one: the frontier of scientific work now sits inside industry.

Sources. Three vintages of the same NSF/NCSES measure (sector of new research doctorate recipients with a definite U.S. employment commitment; postdocs excluded; “industry” includes self-employment): (1) **1970–74** — NSF/SRS (2006), *U.S. Doctorates in the 20th Century*, NSF 06-319, Table 6-3 (five-year average). (2) **1986, 1991** — NORC/NSF, *Doctorate Recipients from U.S. Universities: Summary Report 2006*, Table 30 (selected years 1986–2006). URL: https://www.norc.org/content/dam/norc-org/pdfs/SED_Sum_Rpt_2006.pdf (3) **1994–2024** — NCSES, *Survey of Earned Doctorates 2024*, Table 2-6, NSF 25-349. URL: <https://nces.nsf.gov/pubs/nsf25349/assets/data-tables/tables/nsf25349-tab002-006.pdf> All accessed 2026-06-22. The 1994–2024 segment uses the single current NCSES trend table (no vintage-mixing within it). Per-point sourcing in `data/raw/sed_table2-6_employment_sector.csv` and `data/raw/sed_historical_employment_sector_allfields.csv`.

Exhibit 2 — Founders are not interchangeable managers



Source: Lee, Kim & Bae, "Are Founder CEOs Better Innovators? Evidence from S&P 500 Firms" (Wharton working paper), Figure 2; values digitized from the chart. The peer-reviewed version (Research Policy 49(1), 2020, DOI 10.1016/j.respol.2019.103862) identifies off CEO sudden deaths and reports a ~43.8% drop. Accessed 2026-06-22.

Figure 2: Exhibit 2

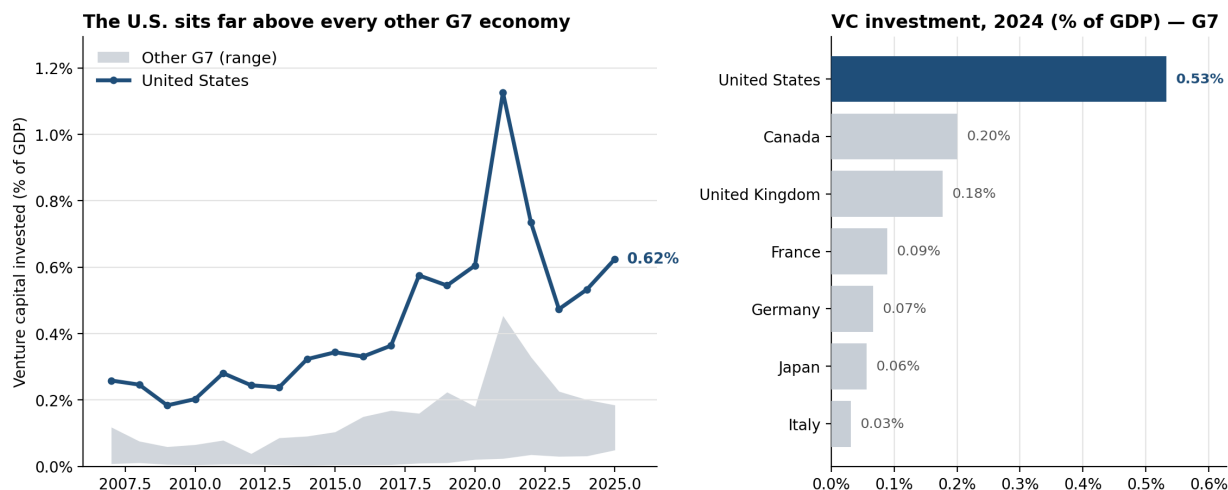
Annotation. This figure speaks to the distinctive skillset a founder brings — and to whether the person who builds a company can be swapped for a professional manager once it is built. It is an event study drawn directly from Lee, Kim, and Bae (2020): when a founder-CEO exits unexpectedly, citation-weighted patents, a standard proxy for innovation, fall. Event studies can be confusing on a first read, because they plot a difference-in-differences rather than a simple time trend. Given more time, I would replicate the authors’ analysis myself so the figure could show the treated and comparison groups as separate lines. **Nuance not to overclaim:** this is just one academic paper, so it should be presented as a finding rather than a fact. Also, I present the authors’ figures; I did not re-run the analysis myself.

Sources. *Event-study figure:* Lee, Joon Mahn; Kim, Jongsoo; Bae, Joonhyung, “Are Founder CEOs Better Innovators? Evidence from S&P 500 Firms” (Wharton Mack Institute working paper; S&P 500, 1993–2003), **Figure 2**, “Switching from Founder CEO to Professional CEO.” Values digitized from the published chart (± 0.03 reading error) into `data/raw/lee_kim_bae_fig2_event_study.csv`. URL: https://mackinstitute.wharton.upenn.edu/wp-content/uploads/2016/03/Mahn-Lee-Joon-Kim-Jongsoo-and-Bae-Joonhyung_Are-Founder-CEOs-Better-Innovators.-Evidence-from-SP-500-Firms.pdf *Headline magnitude (corroboration):* Lee, J.M., Kim, J., & Bae, J. (2020), “Founder CEOs and innovation: Evidence from CEO sudden deaths in public firms,” *Research Policy* 49(1), 103862, DOI 10.1016/j.respol.2019.103862; key estimates in `data/raw/lee_kim_bae_2020_estimates.csv`. *Method:* the figure plots the coefficient-plus-constant from a firm fixed-effects panel OLS of $\ln(1 + \text{citation-weighted patent count})$ on year-relative-to-switch dummies. All accessed 2026-06-22.

Exhibit 3 — Financing the swing: U.S. venture capital dwarfs the rest of the G7

Financing the swing: U.S. venture capital dwarfs the rest of the G7

Venture capital invested as a share of GDP. In 2024 the U.S. invested ~0.5% of GDP — 3x the U.K., 6x France, ~8–9x Germany and Japan.



Source: OECD Entrepreneurship Financing Database, “Venture capital investments (market statistics)”, stage = Total; accessed 2026-06-24. Compared here against the rest of the G7 (the largest advanced economies). Among smaller economies Israel invests a larger share of GDP (~1.8%), but only ~\$10B in absolute terms vs the U.S.’s ~\$156B. VC is one channel of startup finance, not all of it.

Figure 3: Exhibit 3

Annotation. The first two exhibits establish that frontier talent now sits in industry and that founders specifically are hard to replace. This figure attempts to display America’s unique ability to support them. Relative to the size of its economy, the U.S. invests far more than any other G7 country — roughly 0.5% of GDP in 2024, about 3× the U.K., 6× France, and 8–9× Germany and Japan. **Why show peers rather than just China:** the contest with China is usually framed as a two-country race, but particularly as the EU aims to lower its reliance on the United States, it’s worth making the point that the capacity to finance

founders at scale is what determines who can credibly *challenge* Beijing. On that measure, the U.S. stands alone even among its closest allies. This lets us expand the argument slightly. Showing the U.S. against its own peer group demonstrates that America is the one country actually equipped to mount the challenge. **Nuance not to overclaim:** the U.S. is not the world leader on this ratio — *Israel* invests a larger share of its GDP (~1.8%). I scope the comparison to the G7, the large advanced economies where the U.S. leads.

Source. OECD, *Entrepreneurship Financing Database*, “Venture capital investments (market statistics),” dataflow DSD_VC@DF_VC_INV, business development stage = Total. Units used: venture capital as a percentage of GDP, and absolute US\$ (exchange-rate converted) for the scale comparison. Pulled via the OECD SDMX API, accessed 2026-06-24; snapshot cached in `data/raw/oecd_vc_gdp.csv`. *Coverage:* 37 reporting economies, 2002–2025; the figure uses 2024, the latest year with full G7 coverage, for the cross-section and 2006–2024 for the trend. Data Explorer: https://data-explorer.oecd.org/vis?df%5Bid%5D=DSD_VC@DF_VC_INV.

How to reproduce

```
pip install -r requirements.txt
python code/exhibit1_stem_phd_pathways.py
python code/exhibit2_founder_ceo_innovation.py
python code/exhibit3_vc_gdp.py
```

Each script prints the key figures it computes and writes its figure to this folder. Methodological choices are documented at the top of each script and in the annotations above.